

LCA at Tallgrass: One-Page Brief

Oct 2, 2026 · Andrew Gibson

How an LCA works

An LCA counts every greenhouse gas emitted and removed over a product's life. For Trailblazer, it sets each shipper's 45Z credit, its clean fuel credits and its removal-credit yield. Four phases (ISO 14040/44):

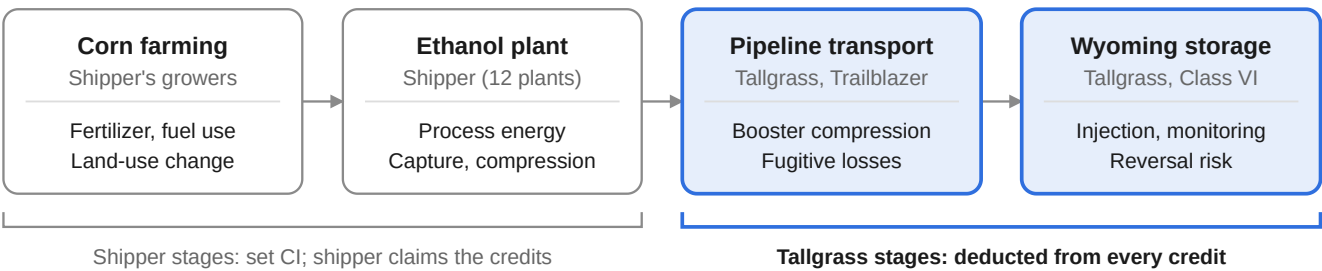
1. **Goal and scope:** what unit, which boundaries (corn field to Wyoming storage).
2. **Inventory:** every input and output, from fertilizer to booster compression power.
3. **Impact:** add it up in CO₂e, minus land-use and supply-chain charges.
4. **Interpretation:** test what drives the answer (land use, grid power, leakage).

One tonne, four rulebooks

Program	Unit	Corn land use	What CCS earns
45Z	g/MJ fuel	Excluded after 2025	About 30 CI points; \$195–260/t to 2029
CA LCFS	g/MJ fuel	Included	CI cut, with a CARB CCS pathway
45Q	Tonne stored	Not counted	\$85/t, metered
Puro / Microsoft	Net tonne removed	Charged (~1.5 kg/gal)	Net credits of ~40–50% of captured tonnes

Where Tallgrass sits

Tallgrass controls the two stages buyers audit most closely



How Tallgrass applies it

- **Own the storage-stage LCA.** One audited data package of metering, compression power and Class VI monitoring that all 12 shippers can cite. Charge a fee or revenue share.
- **Grade tonnes by full LCA.** Sell premium tonnes to strict buyers (Microsoft, Frontier) and standard tonnes to spot buyers.
- **Screen every new CO₂ source.** Coal fails as a substitute for corn: it earns only 45Q at \$85/t, no clean fuel or removal credits, and capture costs roughly \$60–100/t. It may be worth adding only as pipeline volume under a utility take-or-pay contract.

Sources: Trailblazer Low-Carbon Market Strategy; Microsoft 2026 CDR criteria; Puro.earth GSC v.5; 26 USC 45Z and 45Q. Figures marked ~ are estimates.